IMPERIAL COLLEGE LONDON

EE4-18
EE9-AO6

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

RADIO FREQUENCY ELECTRONICS

Friday, 15 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : W.T. Pike

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Special instructions for invigilators

This is a closed book examination.

Special instructions for students

Answer any four (4) out of the six (6) questions.

All variable have their usual meaning.

Standard filter curves and tables are given at the back.

The Questions

1.

a) List five wireless applications that are currently being used within the security and defences industries and briefly describe (in a few sentences) how each application exploits the radio frequency spectrum.

[10]

b) State a well-known example of how an older technology has been updated by adopting RF electronics and explain how this change provides an adaptive and agile solution for modern warfare.

[4]

c) From a technical perspective, explain how RF electronics can be used for crowd control.

[3]

d) From a technical perspective, explain the limitation of autonomous cruise control.

[3]

2.

a) Draw the generic block diagram of a half-duplex transceiver, between IF and RF ports, indicating the directions of signal flow. Clearly identify each block and assign a reasonable value for the power gain or loss typically found for each. Then determine typical power levels at the input and output of each block. All assumptions must be clearly stated; for example, typical input and output power levels.

[10]

b) For the block diagram from 2(a):

i) Briefly explain the problem with both short-term and long-term stability associated with the LO.

[2]

ii) Briefly explain the problem of LO-RF leakage when operating in 'radio silence'.

[2]

iii) Why does the PA need to be driven? Briefly compare and contrast the gain and output power in both stages.

[2]

iv) Briefly explain the problem with poor isolation in the T/R Switch and estimate a typical poor isolation level.

[2]

v) What is the function of a Diversity Switch?

[1]

vi) What is the function of a Limiter?

[1]

3. Consider the 2-element phased array antenna, shown in Figure 3.1, having a mid-band angular frequency of operation ω_o. All variables have their usual meaning.

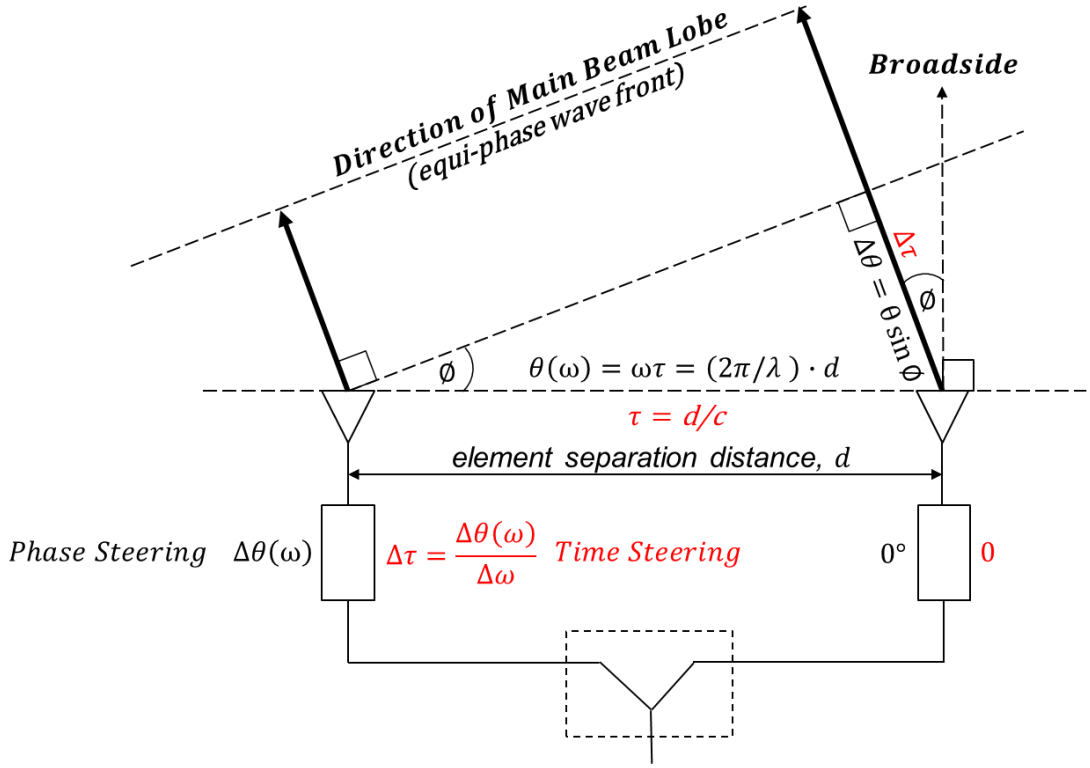


Figure 3.1 Two-element Phased Array Antenna. (θ and $\Delta\theta$ are independent variables)

a) Mathematically define the characteristics of a true phase shifter and illustrate the frequency responses of relevant performance parameters $\Delta\theta(\omega)$ and $\Delta\tau(\omega)$.

[4]

b) If the antenna in Figure 3.1 employs true phase steering, mathematically prove that the direction of the main beam from broadside ϕ is a function of frequency. What is this phenomenon called?

[4]

c) What physical separation distance between elements d is commonly used and why? Also, what practical limitation does this pose when scaling with frequency?

[4]

d) With the separation distance state in 3(c), derive the resulting expression for $\phi(\omega)$.

[3]

e) Using the expression derived in 3(d), calculate $\phi(\omega)$ for a 90° relative phase shift for the following three values of $\omega/\omega_o = [0.9, 1.0, 1.1]$.

[3]

f) If the scenario in 3(e) is implemented by a 4-element phased array antenna, what is the maximum relative phase shift that is needed?

[2]

4. Consider a wireless line-of-sight communications link operating at 76.5 GHz.

a) With the use of a simple sketch (clearly labelled with the appropriate variables), derive Friis' link equation in log form (showing the appropriate units for each variable) for a line-of-sight wireless link. Interconnecting feed line losses must be included.

[6]

b) The gain of an antenna is given in equation (4.1). All variable have their usual meaning.

$$G = \frac{A_e}{A_i} \cdot \eta \quad (4.1)$$

where,
Effective antenna aperture area, A_e
Effective isotropic aperture area, $A_i = \lambda_o^2/(4\pi)$
Antenna efficiency, η

Tx and Rx antennas are identical and made from the open-ended apertures of the associated WR-10 waveguide feed lines, having internal cross-sectional dimensions of 2.54 mm x 1.27 mm. Assuming a 100% antenna efficiency, calculate the effective antenna gain for these apertures.

[2]

c) The feed lines have a wave impedance normalized to that of free space of 1.5716. Calculate:

i) The voltage-wave reflection coefficient at the aperture (boundary between the waveguide and free space) and the resulting return loss. Comment on the significance or otherwise of this return loss.

[2]

ii) Mathematically, define the principle of 'Conservation of Energy', in relation to the S-parameters for a symmetrical and reciprocal passive 2-port network, and then apply this to calculate the antenna efficiency and the impedance mismatch loss at the aperture.

[4]

d) Calculate the minimum combined far-field distance between the Tx and Rx antennas in 4(b).

[2]

e) The Tx power is 30 mW and the identical interconnecting feed lines have an insertion loss of -0.1 dB that are connected to the identical aperture antennas (which are perfectly aligned for optimal line-of-sight communications). Choose any value of range within the far field, between the antennas, and calculate:

i) Path loss.

[2]

ii) Rx power.

[2]

5. Consider the three-port broadband directional coupler shown in Figure 5.1. Holes spaced $\lambda_g/4$ apart allow waves travelling towards Port 3 to combine, and ideally cancel in the opposite direction. A manufacturer has given the following relevant specifications:

Coupling:	10 dB	Directivity (or Isolation):	40 dB
Main Line VSWR:	1.10	Auxiliary Line VSWR:	1.17

a) Stating all assumptions, extract the complex S-parameter matrix $[S^c]$ for this coupler, given in equation 5.1, for this 3-port network using the given manufacturer's specifications.

$$[S^c] = \begin{pmatrix} S_{11}^c & S_{12}^c & S_{13}^c \\ S_{21}^c & S_{22}^c & S_{23}^c \\ S_{31}^c & S_{32}^c & S_{33}^c \end{pmatrix} \quad (5.1) \qquad [5]$$

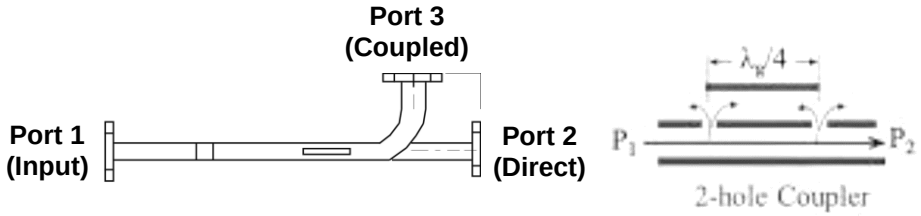


Figure 5.1 Three-port Broadband Directional Coupler.

b) A four-port reflectometer can be used to measure the Return Loss of a Device Under Test (DUT). Connecting power detectors to Ports 2 and 4 of the four-port network shown in Figure 5.2 is one example. Using intuitive reasoning, briefly explain the operation of this circuit for measuring a one-port DUT that presents an arbitrary impedance to Port 3. Briefly explain what happens to the measured Return Loss if too much RF power is generated by the source.

[5]

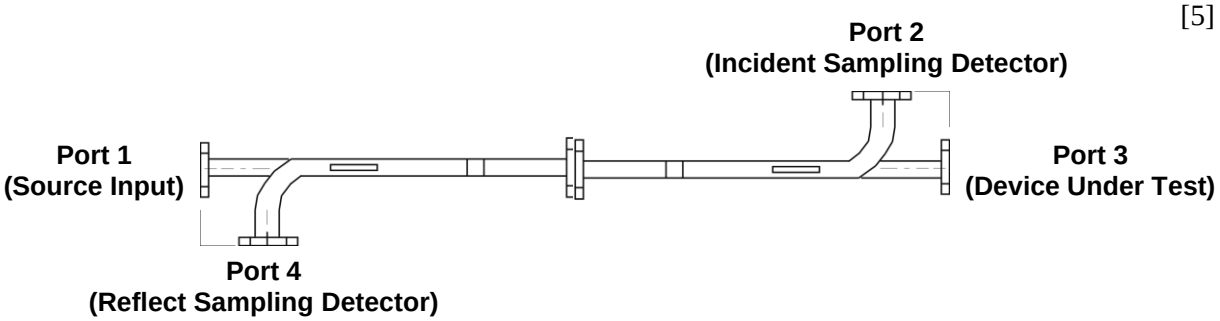


Figure 5.2 Four-port Reflectometer (source, power detectors and DUT not shown).

c) Now apply reasonable simplifying approximations to the S-parameter matrix in 5(a). Hint: if the magnitude of a matrix element is less than 0.1 then it can be set to zero. Using the simplified $[S^c]$ elements and intuitive reasoning, determine the complex S-parameter matrix $[S^R]$, given in equation 5.2, for the four-port reflectometer network with Port 3 terminated by a one-port DUT having an arbitrary voltage-wave reflection coefficient ρ_{DUT}. The source and power detectors can be assumed to have perfect impedance matching.

$$[S^R] = \begin{pmatrix} S_{11} & S_{12} & S_{14} \\ S_{21} & S_{22} & S_{24} \\ S_{41} & S_{42} & S_{44} \end{pmatrix} \quad (5.2) \qquad [5]$$

d) Assume that $S_{21} \approx b_2/a_1$ and $S_{41} \approx b_4/a_1$ in 5(c), where a and b respectively represent voltage waves entering and leaving the network port, and square-law detectors are employed (operating in their linear region) producing corresponding output voltages V_2 and V_4. Determine an expression for the measured Return Loss of the Device Under Test in terms of V_2, V_4 and S_{21}^c. Using S_{21}^c from 5(a), and measured values of V_4 = 0.04 V and V_2 = 1 V, calculate the measured Return Loss.

[5]

6. Design a 3rd order Chebyshev *LC* band-pass filter to meet the following specifications:

Centre frequency of operation: $f_o = 2$ GHz
-3 dB Bandwidth: 20 MHz
Worst-case passband ripple: 0.1 dB
Source and load impedances: $Z_o = 50\ \Omega$
First branch: Series

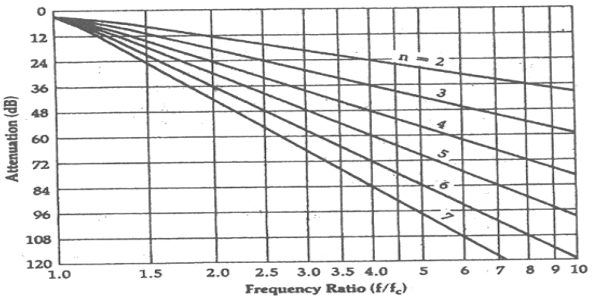
a) Design the filter with the use of series and shunt *LC* resonators.

[10]

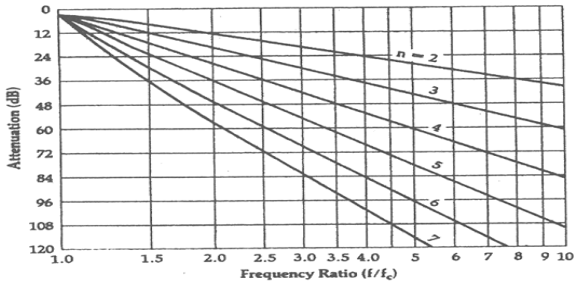
b) Redesign the filter in 6(a) to avoid the use of shunt *LC* resonators by employing inductor-based K-inverters having $K^2 = 1{,}000$.

[10]

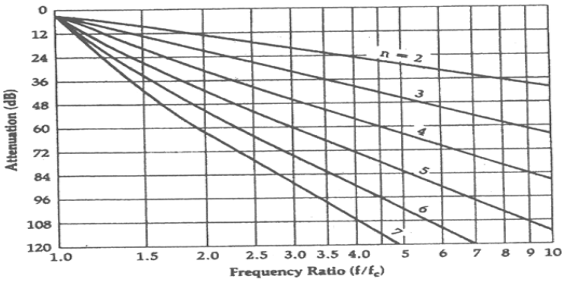
Standard Filter Curves and Tables



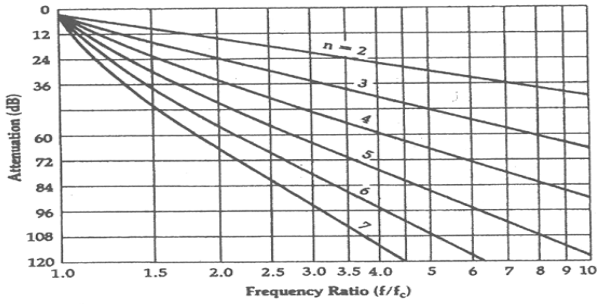
Attenuation characteristics for Butterworth filters.



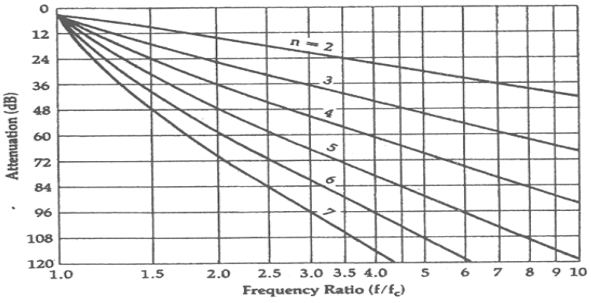
Attenuation characteristics for a Chebyshev filter with 0.01-dB ripple.



Attenuation characteristics for a Chebyshev filter with 0.1-dB ripple.

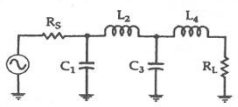


Attenuation characteristics for a Chebyshev filter with 0.5-dB ripple.

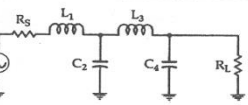


Attenuation characteristics for a Chebyshev filter with 1-dB ripple.

Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

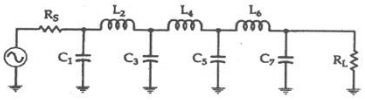


R_S/R_L	C_1	L_2	C_3	L_4
3.000	0.572	3.132		
4.000	0.365	4.600		
8.000	0.157	9.658		
∞	1.213	1.109		
1.000	2.216	1.088	2.216	
0.500	4.431	0.817	2.216	
0.333	6.647	0.726	2.216	
0.250	8.862	0.680	2.216	
0.125	17.725	0.612	2.216	
∞	1.652	1.460	1.108	
3.000	0.653	4.411	0.814	2.535
4.000	0.452	7.083	0.612	2.848
8.000	0.209	17.164	0.428	3.281
∞	1.350	2.010	1.488	1.106

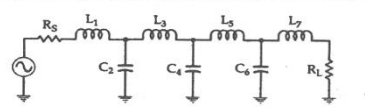


R_L/R_S	L_1	C_2	L_3	C_4

Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

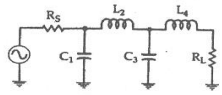


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	2.207	1.128	3.103	1.128	2.207		
	0.500	4.414	0.565	4.653	1.128	2.207		
	0.333	6.622	0.376	6.205	1.128	2.207		
	0.250	8.829	0.282	7.756	1.128	2.207		
	0.125	17.657	0.141	13.961	1.128	2.207		
	∞	1.721	1.645	2.061	1.493	1.103		
6	3.000	0.679	3.873	0.771	4.711	0.969	2.406	
	4.000	0.481	5.644	0.476	7.351	0.849	2.582	
	8.000	0.227	12.310	0.198	16.740	0.726	2.800	
	∞	1.378	2.097	1.690	2.074	1.494	1.102	
7	1.000	2.204	1.131	3.147	1.194	3.147	1.131	2.204
	0.500	4.408	0.566	6.293	0.895	3.147	1.131	2.204
	0.333	6.612	0.377	9.441	0.796	3.147	1.131	2.204
	0.250	8.815	0.283	12.588	0.747	3.147	1.131	2.204
	0.125	17.631	0.141	25.175	0.671	3.147	1.131	2.204
	∞	1.741	1.677	2.155	1.703	2.079	1.494	1.102



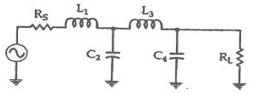
n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

Butterworth Low-Pass Prototype Element Values

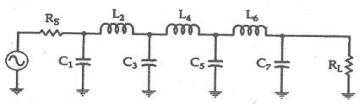


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.111	1.035	1.835		
	1.250	0.849	2.121		
	1.429	0.697	2.439		
	1.667	0.566	2.828		
	2.000	0.448	3.346		
	2.500	0.342	4.095		
	3.333	0.245	5.313		
	5.000	0.156	7.707		
	10.000	0.074	14.814		
	∞	1.414	0.707		
3	0.900	0.808	1.633	1.599	
	0.800	0.844	1.384	1.926	
	0.700	0.915	1.165	2.277	
	0.600	1.023	0.965	2.702	
	0.500	1.181	0.779	3.261	
	0.400	1.425	0.604	4.064	
	0.300	1.838	0.440	5.363	
	0.200	2.669	0.284	7.910	
	0.100	5.167	0.138	15.455	
	∞	1.500	1.333	0.500	
4	1.111	0.466	1.592	1.744	1.469
	1.250	0.388	1.695	1.511	1.811
	1.429	0.325	1.862	1.291	2.175
	1.667	0.269	2.103	1.082	2.613
	2.000	0.218	2.452	0.883	3.187
	2.500	0.169	2.986	0.691	4.009
	3.333	0.124	3.883	0.507	5.338
	5.000	0.080	5.684	0.331	7.940
	10.000	0.039	11.094	0.162	15.642
	∞	1.531	1.577	1.082	0.383

n	R_L/R_S	L_1	C_2	L_3	C_4

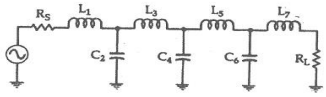


Butterworth Low-Pass Prototype Element Values

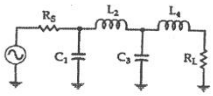


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	0.900	0.442	1.027	1.910	1.756	1.389		
	0.800	0.470	0.866	2.061	1.544	1.738		
	0.700	0.517	0.731	2.285	1.333	2.108		
	0.600	0.586	0.609	2.600	1.126	2.552		
	0.500	0.686	0.496	3.051	0.924	3.133		
	0.400	0.838	0.388	3.736	0.727	3.965		
	0.300	1.094	0.285	4.884	0.537	5.307		
	0.200	1.608	0.186	7.185	0.352	7.935		
	0.100	3.512	0.091	14.095	0.173	15.710		
	∞	1.545	1.694	1.382	0.894	0.309		
6	1.111	0.289	1.040	1.322	2.054	1.744	1.335	
	1.250	0.245	1.116	1.126	2.239	1.550	1.688	
	1.429	0.207	1.236	0.957	2.499	1.346	2.062	
	1.667	0.173	1.407	0.801	2.858	1.143	2.509	
	2.000	0.141	1.653	0.654	3.369	0.942	3.094	
	2.500	0.111	2.028	0.514	4.141	0.745	3.931	
	3.333	0.082	2.656	0.379	5.433	0.552	5.280	
	5.000	0.054	3.917	0.248	8.020	0.363	7.922	
	10.000	0.026	7.705	0.122	15.786	0.179	15.738	
	∞	1.553	1.759	1.553	1.202	0.758	0.259	
7	0.900	0.299	0.711	1.404	1.489	2.125	1.727	1.296
	0.800	0.322	0.606	1.517	1.278	2.334	1.546	1.652
	0.700	0.357	0.515	1.688	1.091	2.618	1.350	2.028
	0.600	0.408	0.432	1.928	0.917	3.005	1.150	2.477
	0.500	0.480	0.354	2.273	0.751	3.553	0.951	3.064
	0.400	0.590	0.278	2.795	0.592	4.380	0.754	3.904
	0.300	0.775	0.206	3.671	0.437	5.761	0.560	5.258
	0.200	1.145	0.135	5.427	0.287	8.526	0.369	7.908
	0.100	2.257	0.067	10.700	0.142	16.822	0.182	15.748
	∞	1.558	1.799	1.659	1.397	1.055	0.656	0.223

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

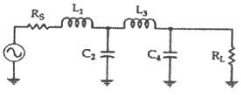


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

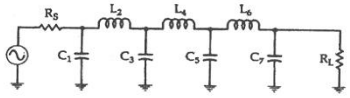


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.101	1.347	1.483		
	1.111	1.247	1.595		
	1.250	0.943	1.997		
	1.429	0.759	2.344		
	1.667	0.609	2.750		
	2.000	0.479	3.277		
	2.500	0.363	4.033		
	3.333	0.259	5.255		
	5.000	0.164	7.650		
	10.000	0.078	14.749		
	∞	1.412	0.742		
3	1.000	1.181	1.821	1.181	
	0.900	1.092	1.660	1.480	
	0.800	1.097	1.443	1.806	
	0.700	1.160	1.228	2.165	
	0.600	1.274	1.024	2.598	
	0.500	1.452	0.829	3.164	
	0.400	1.734	0.645	3.974	
	0.300	2.216	0.470	5.230	
	0.200	3.193	0.305	7.834	
	0.100	6.141	0.148	15.390	
	∞	1.501	1.433	0.591	
4	1.100	0.950	1.938	1.761	1.046
	1.111	0.854	1.946	1.744	1.165
	1.250	0.618	2.075	1.542	1.617
	1.429	0.495	2.279	1.334	2.008
	1.667	0.398	2.571	1.128	2.461
	2.000	0.316	2.994	0.926	3.045
	2.500	0.242	3.641	0.729	3.875
	3.333	0.174	4.727	0.538	5.209
	5.000	0.112	6.910	0.352	7.613
	10.000	0.054	13.469	0.173	15.510
	∞	1.529	1.694	1.312	0.523

n	R_L/R_S	L_1	C_2	L_3	C_4

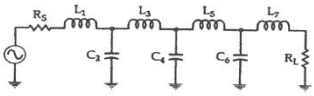


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

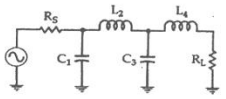


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	0.977	1.685	2.037	1.685	0.977		
	0.900	0.880	1.456	2.174	1.641	1.274		
	0.800	0.877	1.235	2.379	1.499	1.607		
	0.700	0.926	1.040	2.658	1.323	1.977		
	0.600	1.019	0.863	3.041	1.135	2.424		
	0.500	1.166	0.699	3.584	0.942	3.009		
	0.400	1.398	0.544	4.403	0.749	3.845		
	0.300	1.797	0.398	5.772	0.557	5.193		
	0.200	2.604	0.259	8.514	0.368	7.826		
	0.100	5.041	0.127	16.741	0.182	15.613		
	∞	1.547	1.795	1.645	1.237	0.488		
6	1.101	0.851	1.796	1.841	2.027	1.631	0.937	
	1.111	0.760	1.782	1.775	2.094	1.638	1.053	
	1.250	0.545	1.864	1.489	2.403	1.507	1.504	
	1.429	0.436	2.038	1.266	2.735	1.332	1.899	
	1.667	0.351	2.298	1.061	3.167	1.145	2.357	
	2.000	0.279	2.678	0.867	3.768	0.954	2.948	
	2.500	0.214	3.261	0.682	4.667	0.761	3.790	
	3.333	0.155	4.245	0.503	6.163	0.568	5.143	
	5.000	0.100	6.223	0.330	9.151	0.376	7.785	
	10.000	0.048	12.171	0.162	18.105	0.187	15.595	
	∞	1.551	1.847	1.790	1.598	1.190	0.469	
7	1.000	0.913	1.595	2.002	1.870	2.002	1.595	0.913
	0.900	0.816	1.362	2.089	1.722	2.202	1.581	1.206
	0.800	0.811	1.150	2.262	1.525	2.465	1.464	1.538
	0.700	0.857	0.967	2.516	1.323	2.802	1.307	1.910
	0.600	0.943	0.803	2.872	1.124	3.250	1.131	2.359
	0.500	1.080	0.650	3.382	0.928	3.875	0.947	2.948
	0.400	1.297	0.507	4.156	0.735	4.812	0.758	3.790
	0.300	1.669	0.372	5.454	0.546	6.370	0.568	5.148
	0.200	2.242	0.242	8.057	0.360	9.484	0.378	7.802
	0.100	4.701	0.119	15.872	0.178	18.818	0.188	15.652
	∞	1.559	1.867	1.866	1.765	1.563	1.161	0.456

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

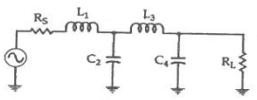


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

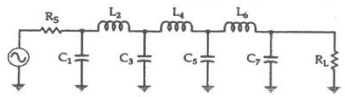


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.355	1.209	1.638		
	1.429	0.977	1.982		
	1.667	0.733	2.489		
	2.000	0.560	3.054		
	2.500	0.417	3.827		
	3.333	0.293	5.050		
	5.000	0.184	7.426		
	10.000	0.087	14.433		
	∞	1.391	0.819		
3	1.000	1.433	1.594	1.433	
	0.900	1.426	1.494	1.622	
	0.800	1.451	1.356	1.871	
	0.700	1.521	1.193	2.190	
	0.600	1.648	1.017	2.603	
	0.500	1.853	0.838	3.159	
	0.400	2.186	0.660	3.968	
	0.300	2.763	0.486	5.279	
	0.200	3.942	0.317	7.850	
	0.100	7.512	0.155	15.466	
	∞	1.513	1.510	0.716	
4	1.355	0.992	2.148	1.585	1.341
	1.429	0.779	2.348	1.429	1.700
	1.667	0.576	2.730	1.185	2.243
	2.000	0.440	3.227	0.967	2.856
	2.500	0.329	3.961	0.760	3.698
	3.333	0.233	5.178	0.560	5.030
	5.000	0.148	7.607	0.367	7.614
	10.000	0.070	14.887	0.180	15.230
	∞	1.511	1.768	1.455	0.673

n	R_L/R_S	L_1	C_2	L_3	C_4

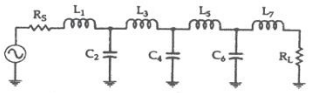


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

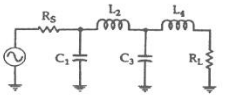


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.301	1.556	2.241	1.556	1.301		
	0.900	1.285	1.433	2.380	1.488	1.488		
	0.800	1.300	1.282	2.582	1.382	1.738		
	0.700	1.358	1.117	2.868	1.244	2.062		
	0.600	1.470	0.947	3.269	1.085	2.484		
	0.500	1.654	0.778	3.845	0.913	3.055		
	0.400	1.954	0.612	4.720	0.733	3.886		
	0.300	2.477	0.451	6.196	0.550	5.237		
	0.200	3.546	0.295	9.127	0.366	7.889		
	0.100	6.787	0.115	17.957	0.182	15.745		
	∞	1.561	1.807	1.766	1.417	0.651		
6	1.355	0.942	2.080	1.659	2.247	1.534	1.277	
	1.429	0.735	2.249	1.454	2.544	1.405	1.629	
	1.667	0.542	2.600	1.183	3.064	1.185	2.174	
	2.000	0.414	3.068	0.958	3.712	0.979	2.794	
	2.500	0.310	3.765	0.749	4.651	0.778	3.645	
	3.333	0.220	4.927	0.551	6.195	0.580	4.996	
	5.000	0.139	7.250	0.361	9.261	0.384	7.618	
	10.000	0.067	14.220	0.178	18.427	0.190	15.350	
	∞	1.534	1.884	1.831	1.749	1.394	0.638	
7	1.000	1.262	1.520	2.239	1.680	2.239	1.520	1.262
	0.900	1.242	1.395	2.361	1.578	2.397	1.459	1.447
	0.800	1.255	1.245	2.548	1.443	2.624	1.362	1.697
	0.700	1.310	1.083	2.819	1.283	2.942	1.233	2.021
	0.600	1.417	0.917	3.205	1.209	3.384	1.081	2.444
	0.500	1.595	0.753	3.764	0.928	4.015	0.914	3.018
	0.400	1.885	0.593	4.618	0.742	4.970	0.738	3.855
	0.300	2.392	0.437	6.054	0.556	6.569	0.557	5.217
	0.200	3.428	0.286	8.937	0.369	9.770	0.372	7.890
	0.100	6.570	0.141	17.603	0.184	19.376	0.186	15.813
	∞	1.575	1.858	1.921	1.827	1.734	1.379	0.631

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

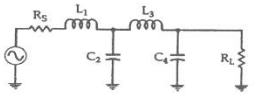


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple

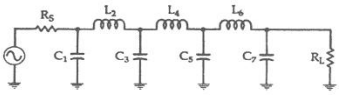


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.984	0.983	1.950		
	2.000	0.909	2.103		
	2.500	0.564	3.165		
	3.333	0.375	4.411		
	5.000	0.228	6.700		
	10.000	0.105	13.322		
	∞	1.307	0.975		
3	1.000	1.864	1.280	1.834	
	0.900	1.918	1.209	2.026	
	0.800	1.997	1.120	2.237	
	0.700	2.114	1.015	2.517	
	0.500	2.557	0.759	3.436	
	0.400	2.985	0.615	4.242	
	0.300	3.729	0.463	5.576	
	0.200	5.254	0.309	8.225	
	0.100	9.890	0.153	16.118	
	∞	1.572	1.518	0.932	
4	1.984	0.920	2.586	1.304	1.826
	2.000	0.845	2.720	1.238	1.985
	2.500	0.516	3.766	0.869	3.121
	3.333	0.344	5.120	0.621	4.480
	5.000	0.210	7.708	0.400	6.987
	10.000	0.098	15.352	0.194	14.262
	∞	1.436	1.889	1.521	0.913

n	R_L/R_S	L_1	C_2	L_3	C_4



Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple



n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.807	1.303	2.691	1.303	1.807		
	0.900	1.854	1.222	2.849	1.238	1.970		
	0.800	1.926	1.126	3.060	1.157	2.185		
	0.700	2.035	1.015	3.353	1.058	2.470		
	0.600	2.200	0.890	3.765	0.942	2.861		
	0.500	2.457	0.754	4.367	0.810	3.414		
	0.400	2.870	0.609	5.296	0.664	4.245		
	0.300	3.588	0.459	6.871	0.508	5.625		
	0.200	5.064	0.306	10.054	0.343	8.367		
	0.100	9.556	0.153	19.647	0.173	16.574		
	∞	1.630	1.740	1.922	1.514	0.903		
6	1.984	0.905	2.577	1.368	2.713	1.299	1.796	
	2.000	0.830	2.704	1.291	2.872	1.237	1.956	
	2.500	0.506	3.722	0.890	4.109	0.881	3.103	
	3.333	0.337	5.055	0.632	5.699	0.635	4.481	
	5.000	0.206	7.615	0.406	8.732	0.412	7.031	
	10.000	0.096	15.186	0.197	17.681	0.202	14.433	
7	1.000	1.790	1.296	2.718	1.385	2.718	1.296	1.790
	0.900	1.835	1.215	2.869	1.308	2.883	1.234	1.953
	0.800	1.905	1.118	3.076	1.215	3.107	1.155	2.168
	0.700	2.011	1.007	3.364	1.105	3.416	1.058	2.455
	0.600	2.174	0.882	3.772	0.979	3.852	0.944	2.848
	0.500	2.428	0.747	4.370	0.838	2.289	0.814	3.405
	0.400	2.835	0.604	5.295	0.685	5.470	0.669	4.243
	0.300	3.546	0.455	6.867	0.522	7.134	0.513	5.635
	0.200	5.007	0.303	10.049	0.352	10.496	0.348	8.404
	0.100	9.458	0.151	19.649	0.178	20.631	0.176	16.665
	∞	1.646	1.777	2.031	1.789	1.924	1.503	0.895

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

